AfyaPlus | Cost-Optimised AI Deployment

EXECUTIVE DECISION BRIEF | OCTOBER 2026 | USD PLANNING ESTIMATES

Conditional GO: human-reviewed pilot

Use the managed API for a small pilot; keep high-risk/uncertain cases on human review. NO-GO for autonomous clinical decisions or production 10x until budget, current rates, latency, and safety gates are evidenced. The runnable local agent is a deterministic clinic-catalog fallback, not a live model call.

Monthly cost scenarios

Scenario	Requests/month	Total / 1,000	Monthly total
Baseline	100,000	$5.3790	$537.90
Sustained 10x	1,000,000	$0.9790	$979.01

At 10x, $979.01 leaves only $20.99 under the proposed $1,000 monthly alert budget. Costs assume 450 input + 180 output tokens and 2% retries; platform and staff allocations are planning estimates, not quotes. Verify current API prices.

Operating controls

Cache: 5/8 public FAQ fixtures hit (62.5%); estimated token cost fell **Cache:** 5/8 sample public questions were served from cache (62.5%); estimated token cost fell from $0.17901 to $0.06713 per 1,000. Cached answers expire after five minutes; never cache patient-specific triage. Fixture matches are not clinical quality evidence. **Batch:** 7 offline jobs grouped into 2 work groups (71.4% fewer); no bulk-pricing discount or throughput gain claimed.

Main risk: a traffic/retry spike can exceed budget and slow service. Mitigate with alerts at $800 actual, $1,000 actual, and $1,000 forecast; cost tags; capped retries/concurrency; automatic stopping after repeated provider failures; and human review. These alerts notify operators; No paid cloud, Redis, GPU, or live 10x test is claimed.

Trade-off and decision gate

Pilot gate: keep interactive p95 under 2 seconds (p95 = 95 of 100 requests finish within 2s), verify live rates, and pass controlled load and safety-quality checks before traffic expansion. Pause on projected budget breach or SLO/quality failure. Keep the local Compose/fixture fallback available.